GPIO Library Installation

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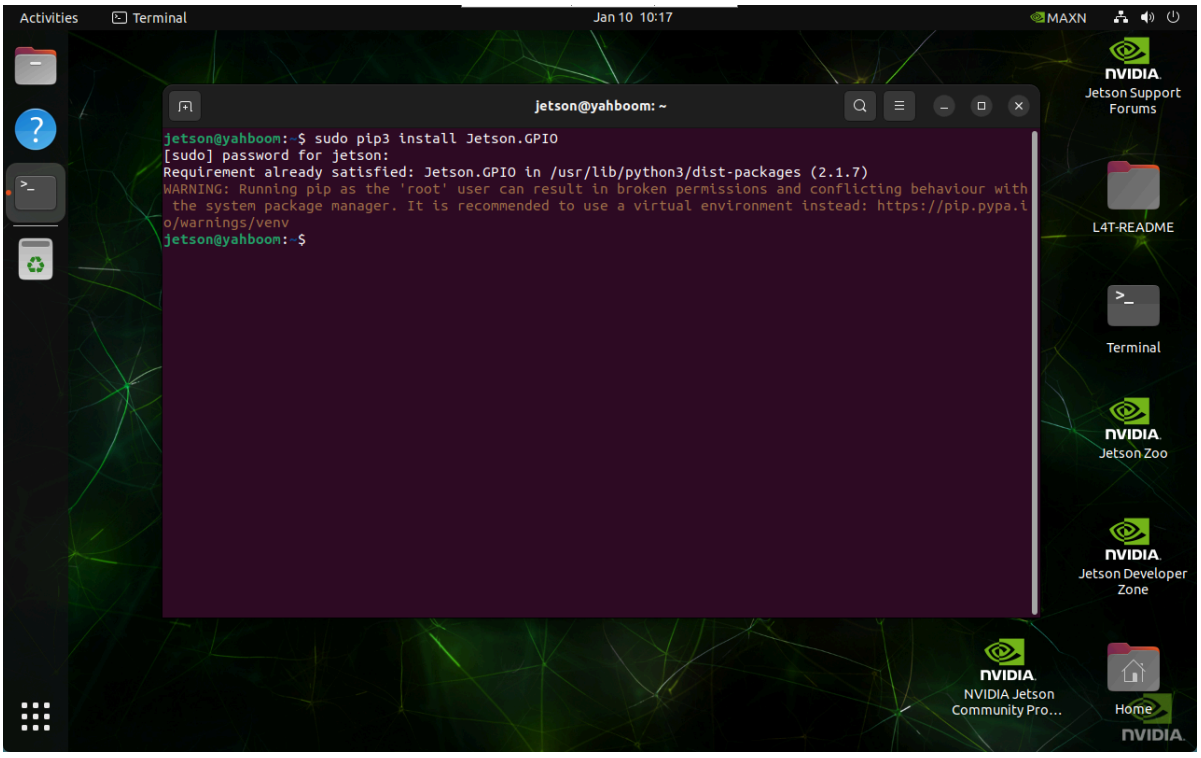
> **pip Installation Note:** Ubuntu 24.04 marks the system Python as "externally managed" according to PEP 668, so installing directly with pip may report `externally-managed-environment`. `--break-system-packages` allows pip to install packages in that environment; a virtual environment usually does not need this parameter, but this tutorial keeps it to keep the commands consistent.

1. Install the Jetson.GPIO Library

The Jetson.GPIO library is installed by default in the system, so this step can be skipped.

1.1. Automatic Installation

```
sudo pip3 install Jetson.GPIO --break-system-packages
```



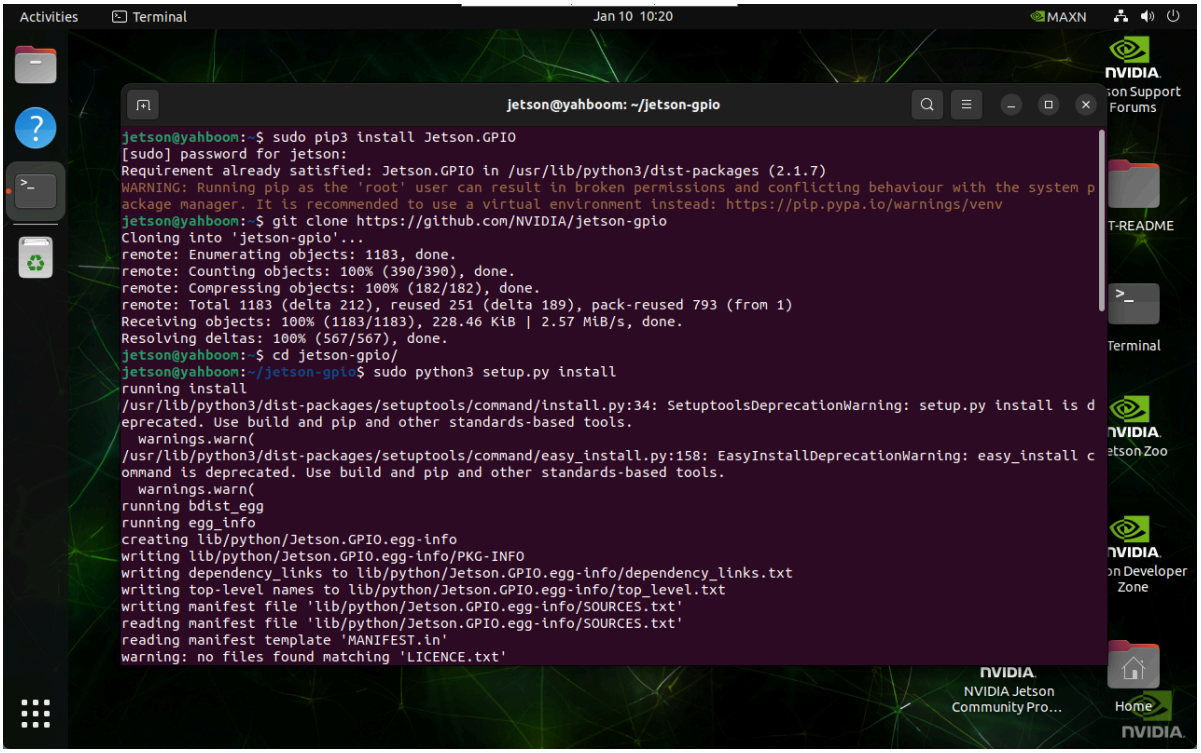
1.2. Manual Installation

The automatic installation method is recommended; a manual installation may not be the latest version.

```
git clone https://github.com/NVIDIA/jetson-gpio
```

```
cd ~/jetson-gpio/
```

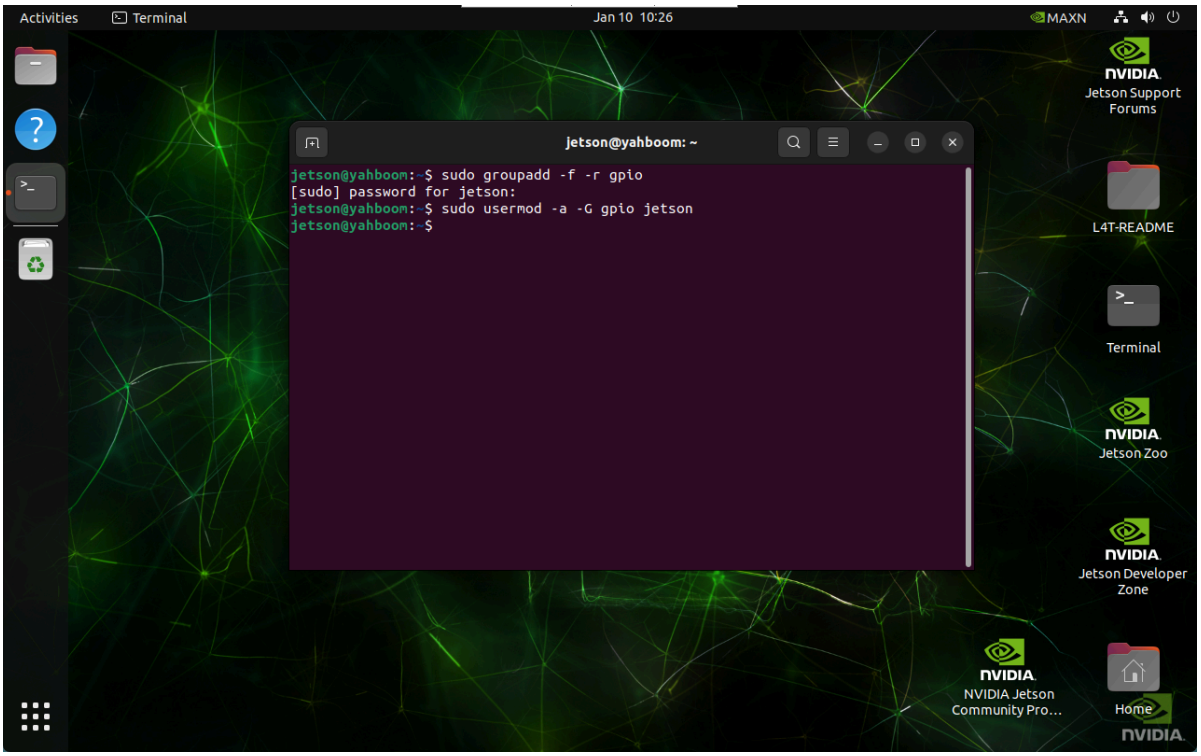
```
sudo python3 setup.py install
```



2. Set User Permissions

Allow the current system user to access and use the Jetson.GPIO library: where `jetson` is the system user name

```
sudo groupadd -f -r gpio
sudo usermod -a -G gpio jetson
```



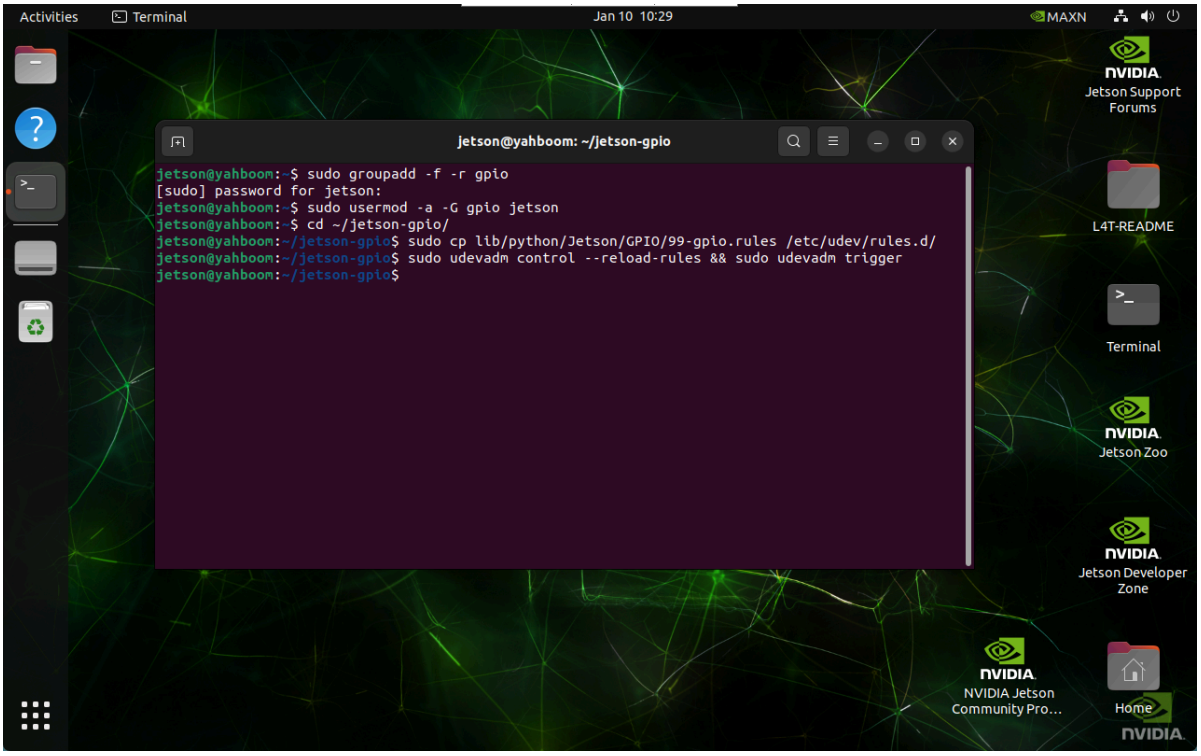
3. Custom Rule File

3.1. Copy the Rule File

```
cd ~/jetson-gpio/
sudo cp lib/python/Jetson/GPIO/99-gpio.rules /etc/udev/rules.d/
```

3.2. Reload udev Rules

```
sudo udevadm control --reload-rules && sudo udevadm trigger
```



4. Set the Mainboard Model

Before controlling GPIO, the mainboard model needs to be set in the terminal:

```
export JETSON_MODEL_NAME=JETSON_ORIN_NANO
```

5. References

https://github.com/NVIDIA/jetson-gpio